CPV301 — Computer Vision
Assignment (Group Project): Satellite Image Analysis of Đà Nẵng with Sentinel-2

- Type: Group project, ongoing across the course
- Weight: 30% of total grade

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1 Why self-collected Sentinel-2 data over Đà Nẵng?

Satellite imagery is a legitimate CV application domain, and Copernicus Sentinel-2 data is free, open, and unambiguously licensed for any use (Copernicus Sentinel Data Terms and Conditions) — no watermark/copyright issue to manage.

Instead of a pre-built dataset, each group collects its own imagery: two Sentinel-2 scenes of the same part of Đà Nẵng, taken on two different dates. This keeps one consistent, meaningful area of study running through all 5 phases below.

1.1 Group assignment: pick a different Đà Nẵng sub-area per group

To avoid every group submitting near-identical AOIs (and to keep cross-group comparison/copying in check), each group should claim a different sub-area, e.g.:

#	Suggested sub-area	Why it's interesting
1	Sơn Trà Peninsula & Hàn River mouth	Forest vs. urban boundary, river delta
2	Mỹ Khê – Non Nước beach corridor	Rapid resort/hotel construction over recent years
3	Hải Châu / Thanh Khê urban core	Dense existing urban fabric, slower change
4	Liên Chiểu port & industrial zone	Port expansion, industrial land reclamation
5	Cẩm Lệ / Hòa Xuân new urban area	Farmland $\rightarrow$ new residential development, high-contrast change

Instructor can assign these or let groups self-select on a first-come basis.

1.2 Data source & collection instructions

1. Register a free account at the Copernicus Data Space Ecosystem Browser (https://dataspace.copernicus.eu) or Sentinel Hub EO Browser.
2. Search Sentinel-2 Level-2A products (already atmospherically corrected) over your assigned AOI.
3. Pick two dates, ideally 2–5 years apart, in the same season/month across years (e.g., March 2019 vs. March 2024) — this isolates real land-cover/urban change from seasonal vegetation variation. Filter for low cloud cover ($<$15–20%) on both dates; Đà Nẵng's dry season (roughly February–August) gives better odds of clear scenes.
4. Export both dates as GeoTIFF (or PNG for true color), including at minimum Red, Green, Blue, and NIR bands (needed for NDVI in Phase 1).
5. Export size limit $\rightarrow$ real stitching case: most free export tools cap a single request's pixel dimensions. If your chosen AOI is large enough to exceed that limit, you'll need to export it in two overlapping pieces and stitch them back together — this is now a genuine constraint, not a contrived one (see Phase 3).

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2 Project Structure — 5 Phases (mapped to when each topic is taught)

Phase	Syllabus topic covered	Target slot	LOs
0 — Setup & Data Collection	Image formation	by slot 28	LO1, LO2
1 — Image Processing	Point ops, filtering	by slot 28	LO3
2 — Segmentation & Features	Feature detection, Segmentation	by slot 36	LO4, LO5
3 — Alignment (+ Stitching if needed)	Feature-based alignment, Image stitching	by slot 42	LO6, LO7
4 — Detection & Change Tracking	Object detection, Recognition	by slot 50	LO8, LO9
5 — Report & Demo	—	by slot 55	all

You don't have to wait for each topic to be lectured before starting — but each phase will make a lot more sense once its corresponding lecture has happened, so this pacing is the recommended one.

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3 Phase 0 — Setup & Data Collection (LO1, LO2)

1. Register on Copernicus Data Space Ecosystem / Sentinel Hub, search, and export your two-date Sentinel-2 pair as described above. Confirm both images load correctly and cover the same geographic extent.
2. Write a short explanation (½ page): how does Sentinel-2 multispectral image formation differ from a normal RGB camera (number/type of bands, spatial resolution $\sim$10–60m/pixel, revisit time, no single "lens focal length" in the traditional sense)? What does each band you plan to use (Red, Green, Blue, NIR) actually measure?

4 Phase 1 — Image Processing (LO3)

1. Compute NDVI = (NIR − Red) / (NIR + Red) for both dates — a point operator applied band-by-band. Visualize as a heatmap and compare the two dates.
2. Apply at least one denoising/smoothing filter (Gaussian or bilateral) to reduce sensor noise, and one contrast-enhancement technique (e.g., histogram equalization) to improve visual interpretability.
3. Briefly justify your filter choices and show before/after comparisons.

5 Phase 2 — Segmentation & Features (LO4, LO5)

1. Detect edges/features in your scene (roads, coastline, riverbanks, field/building boundaries) using an edge/feature detector from the course (Canny, SIFT/ORB keypoints, etc.).
2. Segment your scene into land-cover regions using a technique from the course (mean shift, split-and-merge, or active contours around a specific region of interest — e.g., outlining the beach/coastline or a construction site).
3. Validation: since there's no ground-truth label set here, sanity-check your segmentation by comparing 3–5 regions against a basemap (Google Maps / OpenStreetMap satellite layer) for the same area. Report where your segmentation matches visible land use and where it doesn't, and why.

6 Phase 3 — Alignment (LO6) + Stitching if your AOI required a split export (LO7)

1. Co-registration (required): the two dates of your pair are almost never pixel-aligned (different satellite pass, slight offset). Use feature detection + matching + RANSAC to estimate a homography and align date 2 onto date 1's coordinate frame. This step is required before Phase 4's change tracking will make sense.

2. Stitching: if you had to split your AOI into two overlapping exports in Phase 0 (because it exceeded your tool's single-export size limit), stitch them back into one mosaic here using the same homography-estimation pipeline. If your AOI fit in a single export, this sub-task is optional bonus (+5 pts) — deliberately re-export two overlapping crops of your area to demonstrate the technique.

7 Phase 4 — Detection & Change Tracking (LO8, LO9)

1. Detection: using your Phase 2 segmentation approach (or a small classifier), detect and label regions of a specific class of interest in your scene (e.g., built-up/urban area, water, vegetation, or a specific object type like ships in port for the Liên Chiểu area).
2. Change tracking: using the two co-registered dates from Phase 3, compute a difference map and identify/outline the region(s) that changed the most (e.g., new construction along the beach corridor, port expansion, farmland converted to urban development). Frame this explicitly as the satellite-imagery analogue of "object tracking" — you are following a region of interest across ordered frames, just at a multi-year cadence instead of video frame rate.
3. Discuss at least one likely source of false change detection (cloud/shadow artifacts, seasonal vegetation variation, sun-angle differences, misregistration) and how you tried to control for it.

8 Phase 5 — Report, Demo & Reflection

1. Written report ($\sim$5–8 pages): methodology, results, and a critical discussion of limitations (resolution limits, cloud cover, atmospheric effects, geometric distortion).
2. Group demo/presentation (10 minutes): live or recorded walkthrough of your pipeline and key results — tell the actual story of how your part of Đà Nẵng changed.
3. AI Audit Log (group-level, one consolidated log): every AI tool interaction used anywhere in the project — see format below.
4. Peer evaluation form: each member privately rates their own and teammates' contribution (used to adjust individual grades within the group grade — see rubric).

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9 AI Audit Log (Required Submission)

One shared log for the whole group, `ai_audit_log.md`, listing every AI-assisted interaction across all phases:

#	Phase	Member	Tool	Prompt (summarized)	What the AI returned	What was changed/verified, and why
1	Phase 3	…	…	…	…	…

10 Grading Rubric

Criterion	Weight	Description
Correct execution / technical results	35%	Each phase's code runs and produces correct, verifiable results (correct NDVI, correct alignment, correct segmentation, sensible change map)
Coverage across CLOs	15%	All required tasks across the 5 phases attempted with reasonable depth (Phase 3's Stitching sub-task is required only if the group's AOI needed a split export — otherwise it's bonus)
Conceptual understanding	20%	Report and demo show real understanding of why each technique was chosen (not just "we called this OpenCV function")
AI-assisted skill	10%	Judged from the AI Audit Log: precise prompts, evidence of catching/fixing AI errors, adapting AI output to satellite data specifics (multi-band images, geospatial alignment) rather than generic RGB-photo code
Critical thinking	20%	Quality of the limitations discussion (Phase 0.3, Phase 4.3, and final report) — recognizing where the pipeline could fail and why